```c
#include <stdio.h>
#include <stdlib.h>
#pragma warning(disable:4996)

//Function Declaration
void languageOption();
void english();
void malay();
void staffEnglish();
void staffMalay();
void studentEnglish();
void studentMalay();
void invalid();
void displayGradeScale();

//Language (Extra Feature)
int language;

//Formula Variable
double GPAS1, GPAS2, GPAS3;
double CGPAS1, CGPAS2, CGPAS3;
double GRADE_POINTS1, GRADE_POINTS2,GRADE_POINTS3;
double TOTAL_QUALITY_POINTS1, TOTAL_QUALITY_POINTS2, TOTAL_QUALITY_POINTS3;
int TOTAL_CREDIT_HOURSS1, TOTAL_CREDIT_HOURSS2, TOTAL_CREDIT_HOURSS3;

//Course Variable
int CREDIT_HOURS;
int SEMESTER;
int noOfCourse, courseAction;
char COURSE_CODE[30], GRADE[10];

//Student'details Variable
char NAME[30];
char ID[10];

//Mode Variable
int IDENTITY;
int action = 0;

//Grade Scale & Language Option
void main()
{
    displayGradeScale();
    printf("\n\n");

    do {
        languageOption();
        if (language == 1)
```

```c
        {
            english();
            if (IDENTITY == 1) {
                staffEnglish();
            }
            else if (IDENTITY == 2)
                studentEnglish();
            else if (IDENTITY == 3)
                system("pause");
            else
                invalid();
        }

        else if (language == 2)
        {
            malay();
            if (IDENTITY == 1)
                staffMalay();
            else if (IDENTITY == 2)
                studentMalay();
            else if (IDENTITY == 3)
                system("pause");
            else
                invalid();
        }

        else if (language == 3)
            return 0;

        else
            invalid();

        printf("\nAny other action? Select 1/2\n1. Yes\n2. No\n");
        scanf("%d", &action);

    } while (action == 1);

    printf("\n");
    printf("Thank you for using the GPA/CGPA calculator.\n");
    return;

}

void languageOption()
{
    printf("1. English\n");
    printf("2. Bahasa Melayu\n");
    printf("3. Exit\n");
    printf("Select your language : ");
```

```
 98      scanf("%d", &language);
 99      return 0;
100  }
101
102  void english()
103  {
104      printf("\n\n");
105      printf("-----------------------------------------------------\n");
106      printf("---[WELCOME TO KOLEJ PASAR GPA/CGPA CALCULATOR]---\n");
107      printf("-----------------------------------------------------\n");
108      printf("\n");
109      printf("1. Staff Administration\n");
110      printf("2. Student\n");
111      printf("3. Exit\n");
112      printf("\n");
113      printf("Enter  1 / 2 to select your identity : ");
114      scanf("%d", &IDENTITY);
115      return 0;
116  }
117
118  void malay()
119  {
120      printf("\n\n");
121      printf("-------------------------------------------------------------\n");
122      printf("---[SELAMAT DATANG KE KOLEJ PASAR GPA/CGPA KALKULATOR]---\n");
123      printf("-------------------------------------------------------------\n");
124      printf("\n");
125      printf("1. Kakitangan Kolej Pasar\n");
126      printf("2. Mahasiswa\n");
127      printf("3. Keluar\n");
128      printf("\n");
129      printf("Masuk 1/2 untuk memilih identiti : ");
130      scanf("%d", &IDENTITY);
131      return 0;
132  }
133
134  void staffEnglish()
135  {
136          printf("\n\n");
137          printf("Enter student\'s name : ");
138          rewind(stdin);
139          scanf("%30[^\n]", NAME);
140          printf("Enter student ID : ");
141          scanf("%10s", ID);
142      do {
143          printf("Enter semester session ( 0 to return ) : ");
144          rewind(stdin);
145          scanf("%d", &SEMESTER);
146
```

```
147         if (SEMESTER == 1) {
148             printf("Enter number of courses taken : "); //Let user to
                  enter number of courses taken
149             scanf("%d", &noOfCourse);
150             for (courseAction = 0; courseAction < noOfCourse; courseAction
                  ++)
151             {
152                 printf("Enter course code %d : ", courseAction + 1);
153                 rewind(stdin);
154                 scanf("%8s", COURSE_CODE);
155                 printf("Grade Obtained : ");
156                 rewind(stdin);
157                 scanf("%s", GRADE);
158                 printf("Credit Hours : ");
159                 rewind(stdin);
160                 scanf("%d", &CREDIT_HOURS);
161 
162                 if (strcmp(GRADE, "A") == 0 || strcmp(GRADE, "a") == 0)
163                     GRADE_POINTS1 = 4.00;
164 
165                 else if (strcmp(GRADE, "A-") == 0 || strcmp(GRADE, "A-")
                      == 0)
166                     GRADE_POINTS1 = 3.75;
167 
168                 else if (strcmp(GRADE, "B+") == 0 || strcmp(GRADE, "b+")
                      == 0)
169                     GRADE_POINTS1 = 3.50;
170 
171                 else if (strcmp(GRADE, "B") == 0 || strcmp(GRADE, "b") ==
                      0)
172                     GRADE_POINTS1 = 3.00;
173 
174                 else if (strcmp(GRADE, "B-") == 0 || strcmp(GRADE, "b-")
                      == 0)
175                     GRADE_POINTS1 = 2.75;
176 
177                 else if (strcmp(GRADE, "C+") == 0 || strcmp(GRADE, "c+")
                      == 0)
178                     GRADE_POINTS1 = 2.50;
179 
180                 else if (strcmp(GRADE, "C") == 0 || strcmp(GRADE, "c") ==
                      0)
181                     GRADE_POINTS1 = 2.00;
182 
183                 else
184                     GRADE_POINTS1 = 0.00;
185 
186                 //Formula
187                 TOTAL_QUALITY_POINTS1 += GRADE_POINTS1 * CREDIT_HOURS;
```

```
188                 TOTAL_CREDIT_HOURSS1 += CREDIT_HOURS;
189             }
190             GPAS1 = (double)TOTAL_QUALITY_POINTS1 / TOTAL_CREDIT_HOURSS1;
191             CGPAS1 = (double)GPAS1;
192 
193             //Result
194             printf("\n\n");
195             printf
                  ("============================================================
                  =\n");
196             printf("Student Name                   : %s\n", NAME);
197             printf("Student ID                     : %s\n", ID);
198             printf("Quality Points Earned          : %.2f\n",
                  TOTAL_QUALITY_POINTS1);
199             printf("Total Quality Points Earned    : %.2f\n",
                  TOTAL_QUALITY_POINTS1);
200             printf("Credit Hours Earned            : %d\n",
                  TOTAL_CREDIT_HOURSS1);
201             printf("Total Credit Hours Earned      : %d\n",
                  TOTAL_CREDIT_HOURSS1);
202             printf("Student's GPA                  : %.2f\n", GPAS1);
203             printf("Student's CGPA                 : %.2f\n", CGPAS1);
204             printf
                  ("============================================================
                  ==\n\n");
205         }
206 
207         else if (SEMESTER == 2) {
208             printf("Enter number of courses taken : "); //Let user to
                  enter number of courses taken
209             scanf("%d", &noOfCourse);
210             for (courseAction = 0; courseAction < noOfCourse; courseAction
                  ++)
211             {
212                 printf("Enter course code %d : ", courseAction + 1);
213                 rewind(stdin);
214                 scanf("%8s", COURSE_CODE);
215                 printf("Grade Obtained : ");
216                 rewind(stdin);
217                 scanf("%s", GRADE);
218                 printf("Credit Hours : ");
219                 rewind(stdin);
220                 scanf("%d", &CREDIT_HOURS);
221 
222                 if (strcmp(GRADE, "A") == 0 || strcmp(GRADE, "a") == 0)
223                     GRADE_POINTS2 = 4.00;
224 
225                 else if (strcmp(GRADE, "A-") == 0 || strcmp(GRADE, "A-")
                    == 0)
```

```
226                     GRADE_POINTS2 = 3.75;
227
228                 else if (strcmp(GRADE, "B+") == 0 || strcmp(GRADE, "b+")
                      == 0)
229                     GRADE_POINTS2 = 3.50;
230
231                 else if (strcmp(GRADE, "B") == 0 || strcmp(GRADE, "b") ==
                      0)
232                     GRADE_POINTS2 = 3.00;
233
234                 else if (strcmp(GRADE, "B-") == 0 || strcmp(GRADE, "b-")
                      == 0)
235                     GRADE_POINTS2 = 2.75;
236
237                 else if (strcmp(GRADE, "C+") == 0 || strcmp(GRADE, "c+")
                      == 0)
238                     GRADE_POINTS2 = 2.50;
239
240                 else if (strcmp(GRADE, "C") == 0 || strcmp(GRADE, "c") ==
                      0)
241                     GRADE_POINTS2 = 2.00;
242
243                 else
244                     GRADE_POINTS2 = 0.00;
245
246             //Formula
247             TOTAL_QUALITY_POINTS2 += GRADE_POINTS2 * CREDIT_HOURS;
248             TOTAL_CREDIT_HOURSS2 += CREDIT_HOURS;
249             }
250             GPAS2 = (double)TOTAL_QUALITY_POINTS2 / TOTAL_CREDIT_HOURSS2;
251             CGPAS2 = (double)(TOTAL_QUALITY_POINTS1 +
                  TOTAL_QUALITY_POINTS2) / (TOTAL_CREDIT_HOURSS1 +
                  TOTAL_CREDIT_HOURSS2);
252
253             //Result
254             printf("\n\n");
255             printf
                  ("============================================================
                  =\n");
256             printf("Student Name                  : %s\n", NAME);
257             printf("Student ID                    : %s\n", ID);
258             printf("Quality Points Earned         : %.2f\n",
                  TOTAL_QUALITY_POINTS2);
259             printf("Total Quality Points Earned   : %.2f\n",
                  TOTAL_QUALITY_POINTS1 + TOTAL_QUALITY_POINTS2);
260             printf("Credit Hours Earned           : %d\n",
                  TOTAL_CREDIT_HOURSS2);
261             printf("Total Credit Hours Earned     : %d\n",
                  TOTAL_CREDIT_HOURSS1 + TOTAL_CREDIT_HOURSS2);
```

```
262             printf("Student's GPA                  : %.2f\n", GPAS2);
263             printf("Student's CGPA                 : %.2f\n", CGPAS2);
264             printf
                  ("============================================================
                  ==\n\n");
265         }
266
267         else if (SEMESTER == 3) {
268             printf("Enter number of courses taken : "); //Let user to
                  enter number of courses taken
269             scanf("%d", &noOfCourse);
270             for (courseAction = 0; courseAction < noOfCourse; courseAction
                  ++)
271             {
272                 printf("Enter course code %d : ", courseAction + 1);
273                 rewind(stdin);
274                 scanf("%8s", COURSE_CODE);
275                 printf("Grade Obtained : ");
276                 rewind(stdin);
277                 scanf("%s", GRADE);
278                 printf("Credit Hours : ");
279                 rewind(stdin);
280                 scanf("%d", &CREDIT_HOURS);
281
282                 if (strcmp(GRADE, "A") == 0 || strcmp(GRADE, "a") == 0)
283                     GRADE_POINTS3 = 4.00;
284
285                 else if (strcmp(GRADE, "A-") == 0 || strcmp(GRADE, "A-")
                      == 0)
286                     GRADE_POINTS3 = 3.75;
287
288                 else if (strcmp(GRADE, "B+") == 0 || strcmp(GRADE, "b+")
                      == 0)
289                     GRADE_POINTS3 = 3.50;
290
291                 else if (strcmp(GRADE, "B") == 0 || strcmp(GRADE, "b") ==
                      0)
292                     GRADE_POINTS3 = 3.00;
293
294                 else if (strcmp(GRADE, "B-") == 0 || strcmp(GRADE, "b-")
                      == 0)
295                     GRADE_POINTS3 = 2.75;
296
297                 else if (strcmp(GRADE, "C+") == 0 || strcmp(GRADE, "c+")
                      == 0)
298                     GRADE_POINTS3 = 2.50;
299
300                 else if (strcmp(GRADE, "C") == 0 || strcmp(GRADE, "c") ==
                      0)
```

```
301                     GRADE_POINTS3 = 2.00;
302
303                 else
304                     GRADE_POINTS3 = 0.00;
305
306                 //Formula
307                 TOTAL_QUALITY_POINTS3 += GRADE_POINTS3 * CREDIT_HOURS;
308                 TOTAL_CREDIT_HOURSS3 += CREDIT_HOURS;
309             }
310             GPAS3 = (double)TOTAL_QUALITY_POINTS3 / TOTAL_CREDIT_HOURSS3;
311             CGPAS3 = (double)(TOTAL_QUALITY_POINTS1 +                          ↵
                  TOTAL_QUALITY_POINTS2 + TOTAL_QUALITY_POINTS3) /              ↵
                  (TOTAL_CREDIT_HOURSS1 + TOTAL_CREDIT_HOURSS2 +                ↵
                  TOTAL_CREDIT_HOURSS3);
312
313             //Result
314             printf("\n\n");
315             printf                                                          ↵
                  ("============================================================ ↵
                  =\n");
316             printf("Student Name                   : %s\n", NAME);
317             printf("Student ID                     : %s\n", ID);
318             printf("Quality Points Earned          : %.2f\n",              ↵
                  TOTAL_QUALITY_POINTS3);
319             printf("Total Quality Points Earned    : %.2f\n",              ↵
                  TOTAL_QUALITY_POINTS1 + TOTAL_QUALITY_POINTS2 +               ↵
                  TOTAL_QUALITY_POINTS3);
320             printf("Credit Hours Earned            : %d\n",                ↵
                  TOTAL_CREDIT_HOURSS3);
321             printf("Total Credit Hours Earned      : %d\n",                ↵
                  TOTAL_CREDIT_HOURSS1 + TOTAL_CREDIT_HOURSS2 +                 ↵
                  TOTAL_CREDIT_HOURSS3);
322             printf("Student's GPA                  : %.2f\n", GPAS3);
323             printf("Student's CGPA                 : %.2f\n", CGPAS3);
324             printf                                                          ↵
                  ("============================================================ ↵
                  ==\n\n");
325         }
326
327         else {
328             printf("Invalid semester\n");
329             printf("\n");
330         }
331
332     } while (SEMESTER != 0);
333     return 0;
334 }
335
336 void staffMalay()
```

```
337 {
338     printf("\n\n");
339     printf("Masuk nama murid : ");
340     rewind(stdin);
341     scanf("%30[^\n]", NAME);
342     printf("Masuk ID murid : ");
343     scanf("%10s", ID);
344     do {
345         printf("Masuk sesi semester ( 0 untuk keluar ): ");
346         rewind(stdin);
347         scanf("%d", &SEMESTER);
348 
349         if (SEMESTER == 1) {
350             printf("Masuk bilangan kursus diambil : "); //Let user to
                  enter number of courses taken
351             scanf("%d", &noOfCourse);
352             for (courseAction = 0; courseAction < noOfCourse; courseAction
                  ++)
353             {
354                 printf("Masuk Kod Kursus %d : ", courseAction + 1);
355                 rewind(stdin);
356                 scanf("%8s", COURSE_CODE);
357                 printf("Gred diperolehi : ");
358                 rewind(stdin);
359                 scanf("%s", GRADE);
360                 printf("Jam Kredit : ");
361                 rewind(stdin);
362                 scanf("%d", &CREDIT_HOURS);
363 
364                 if (strcmp(GRADE, "A") == 0 || strcmp(GRADE, "a") == 0)
365                     GRADE_POINTS1 = 4.00;
366 
367                 else if (strcmp(GRADE, "A-") == 0 || strcmp(GRADE, "A-")
                    == 0)
368                     GRADE_POINTS1 = 3.75;
369 
370                 else if (strcmp(GRADE, "B+") == 0 || strcmp(GRADE, "b+")
                    == 0)
371                     GRADE_POINTS1 = 3.50;
372 
373                 else if (strcmp(GRADE, "B") == 0 || strcmp(GRADE, "b") ==
                    0)
374                     GRADE_POINTS1 = 3.00;
375 
376                 else if (strcmp(GRADE, "B-") == 0 || strcmp(GRADE, "b-")
                    == 0)
377                     GRADE_POINTS1 = 2.75;
378 
379                 else if (strcmp(GRADE, "C+") == 0 || strcmp(GRADE, "c+")
```

```
                    == 0)
380                     GRADE_POINTS1 = 2.50;
381
382                 else if (strcmp(GRADE, "C") == 0 || strcmp(GRADE, "c") ==
                      0)
383                     GRADE_POINTS1 = 2.00;
384
385                 else
386                     GRADE_POINTS1 = 0.00;
387
388                 //Formula
389                 TOTAL_QUALITY_POINTS1 += GRADE_POINTS1 * CREDIT_HOURS;
390                 TOTAL_CREDIT_HOURSS1 += CREDIT_HOURS;
391             }
392
393             GPAS1 = (double)TOTAL_QUALITY_POINTS1 / TOTAL_CREDIT_HOURSS1;
394             CGPAS1 = (double)GPAS1;
395
396             //Result
397             printf("\n\n");
398             printf
                  ("============================================================
                  =\n");
399             printf("Nama murid                      : %s\n", NAME);
400             printf("ID murid                        : %s\n", ID);
401             printf("Point Kualiti diperolehi        : %.2f",
                  TOTAL_QUALITY_POINTS1);
402             printf("Jumlah Point Kualiti diperolehi : %.2f",
                  TOTAL_QUALITY_POINTS1);
403             printf("Jam kredit diperolehi           : %d\n",
                  TOTAL_CREDIT_HOURSS1);
404             printf("Jumlah jam kredit diperolehi    : %d\n",
                  TOTAL_CREDIT_HOURSS1);
405             printf("GPA murid                       : %.2f\n", GPAS1);
406             printf("CGPA murid                      : %.2f\n", CGPAS1);
407             printf
                  ("============================================================
                  ==\n\n");
408         }
409
410         else if (SEMESTER == 2) {
411             printf("Masuk bilangan kursus diambil : "); //Let user to
                  enter number of courses taken
412             scanf("%d", &noOfCourse);
413             for (courseAction = 0; courseAction < noOfCourse; courseAction
                  ++)
414             {
415                 printf("Masuk Kod Kursus %d : ", courseAction + 1);
416                 rewind(stdin);
```

```
417                 scanf("%8s", COURSE_CODE);
418                 printf("Gred diperolehi : ");
419                 rewind(stdin);
420                 scanf("%s", GRADE);
421                 printf("Jam Kredit : ");
422                 rewind(stdin);
423                 scanf("%d", &CREDIT_HOURS);
424 
425                 if (strcmp(GRADE, "A") == 0 || strcmp(GRADE, "a") == 0)
426                     GRADE_POINTS2 = 4.00;
427 
428                 else if (strcmp(GRADE, "A-") == 0 || strcmp(GRADE, "A-")
                      == 0)
429                     GRADE_POINTS2 = 3.75;
430 
431                 else if (strcmp(GRADE, "B+") == 0 || strcmp(GRADE, "b+")
                      == 0)
432                     GRADE_POINTS2 = 3.50;
433 
434                 else if (strcmp(GRADE, "B") == 0 || strcmp(GRADE, "b") ==
                      0)
435                     GRADE_POINTS2 = 3.00;
436 
437                 else if (strcmp(GRADE, "B-") == 0 || strcmp(GRADE, "b-")
                      == 0)
438                     GRADE_POINTS2 = 2.75;
439 
440                 else if (strcmp(GRADE, "C+") == 0 || strcmp(GRADE, "c+")
                      == 0)
441                     GRADE_POINTS2 = 2.50;
442 
443                 else if (strcmp(GRADE, "C") == 0 || strcmp(GRADE, "c") ==
                      0)
444                     GRADE_POINTS2 = 2.00;
445 
446                 else
447                     GRADE_POINTS2 = 0.00;
448 
449                 //Formula
450                 TOTAL_QUALITY_POINTS2 += GRADE_POINTS2 * CREDIT_HOURS;
451                 TOTAL_CREDIT_HOURSS2 += CREDIT_HOURS;
452             }
453             GPAS2 = (double)TOTAL_QUALITY_POINTS2 / TOTAL_CREDIT_HOURSS2;
454             CGPAS2 = (double)(TOTAL_QUALITY_POINTS1 +
                  TOTAL_QUALITY_POINTS2) / (TOTAL_CREDIT_HOURSS1 +
                  TOTAL_CREDIT_HOURSS2);
455 
456             //Result
457             printf("\n\n");
```

```
458             printf
                  ("============================================================
                  =\n");
459             printf("Nama murid                      : %s\n", NAME);
460             printf("ID murid                        : %s\n", ID);
461             printf("Point Kualiti diperolehi        : %.2f",
                  TOTAL_QUALITY_POINTS2);
462             printf("Jumlah Point Kualiti diperolehi : %.2f",
                  TOTAL_QUALITY_POINTS1 + TOTAL_QUALITY_POINTS2);
463             printf("Jam kredit diperolehi           : %d\n",
                  TOTAL_CREDIT_HOURSS2);
464             printf("Jumlah jam kredit diperolehi    : %d\n",
                  TOTAL_CREDIT_HOURSS1 + TOTAL_CREDIT_HOURSS2);
465             printf("GPA murid                       : %.2f\n", GPAS2);
466             printf("CGPA murid                      : %.2f\n", CGPAS2);
467             printf
                  ("============================================================
                  ==\n\n");
468         }
469
470         else if (SEMESTER == 3) {
471             printf("Masuk bilangan kursus diambil : "); //Let user to
                  enter number of courses taken
472             scanf("%d", &noOfCourse);
473             for (courseAction = 0; courseAction < noOfCourse; courseAction
                  ++)
474             {
475                 printf("Masuk Kod Kursus %d : ", courseAction + 1);
476                 rewind(stdin);
477                 scanf("%8s", COURSE_CODE);
478                 printf("Gred diperolehi : ");
479                 rewind(stdin);
480                 scanf("%s", GRADE);
481                 printf("Jam Kredit : ");
482                 rewind(stdin);
483                 scanf("%d", &CREDIT_HOURS);
484
485                 if (strcmp(GRADE, "A") == 0 || strcmp(GRADE, "a") == 0)
486                     GRADE_POINTS3 = 4.00;
487
488                 else if (strcmp(GRADE, "A-") == 0 || strcmp(GRADE, "A-")
                    == 0)
489                     GRADE_POINTS3 = 3.75;
490
491                 else if (strcmp(GRADE, "B+") == 0 || strcmp(GRADE, "b+")
                    == 0)
492                     GRADE_POINTS3 = 3.50;
493
494                 else if (strcmp(GRADE, "B") == 0 || strcmp(GRADE, "b") ==
```

```
                    0)
495                   GRADE_POINTS3 = 3.00;
496
497               else if (strcmp(GRADE, "B-") == 0 || strcmp(GRADE, "b-")
                    == 0)
498                   GRADE_POINTS3 = 2.75;
499
500               else if (strcmp(GRADE, "C+") == 0 || strcmp(GRADE, "c+")
                    == 0)
501                   GRADE_POINTS3 = 2.50;
502
503               else if (strcmp(GRADE, "C") == 0 || strcmp(GRADE, "c") ==
                    0)
504                   GRADE_POINTS3 = 2.00;
505
506               else
507                   GRADE_POINTS3 = 0.00;
508
509               //Formula
510               TOTAL_QUALITY_POINTS3 += GRADE_POINTS3 * CREDIT_HOURS;
511               TOTAL_CREDIT_HOURSS3 += CREDIT_HOURS;
512           }
513           GPAS3 = (double)TOTAL_QUALITY_POINTS3 / TOTAL_CREDIT_HOURSS3;
514           CGPAS3 = (double)(TOTAL_QUALITY_POINTS1 +
                TOTAL_QUALITY_POINTS2 + TOTAL_QUALITY_POINTS3) /
                (TOTAL_CREDIT_HOURSS1 + TOTAL_CREDIT_HOURSS2 +
                TOTAL_CREDIT_HOURSS3);
515
516           //Result
517           printf("\n\n");
518           printf
                ("============================================================
                =\n");
519           printf("Nama murid                     : %s\n", NAME);
520           printf("ID murid                       : %s\n", ID);
521           printf("Point Kualiti diperolehi       : %.2f",
                TOTAL_QUALITY_POINTS3);
522           printf("Jumlah Point Kualiti diperolehi : %.2f",
                TOTAL_QUALITY_POINTS1 + TOTAL_QUALITY_POINTS2 +
                TOTAL_QUALITY_POINTS3);
523           printf("Jam kredit diperolehi          : %d\n",
                TOTAL_CREDIT_HOURSS3);
524           printf("Jumlah jam kredit diperolehi    : %d\n",
                TOTAL_CREDIT_HOURSS1 + TOTAL_CREDIT_HOURSS2 +
                TOTAL_CREDIT_HOURSS3);
525           printf("GPA murid                      : %.2f\n", GPAS3);
526           printf("CGPA murid                     : %.2f\n", CGPAS3);
527           printf
                ("============================================================
```

```
                ==\n\n");
528         }
529
530         else {
531             printf("Sesi semester tidak sah");
532             printf("\n");
533         }
534
535     } while (SEMESTER != 0);
536     return 0;
537 }
538
539 void studentEnglish()
540 {
541     printf("Enter Student ID to log in : ");
542     scanf("%s", ID);
543     do {
544         printf("Enter which semester session ( 0 to exit ) : ");
545         scanf("%d", &SEMESTER);
546         if (SEMESTER == 1)
547         {
548             printf("\n\n");
549             printf                                                    ↵
                  ("========================================================= ↵
                  =\n");
550             printf("Student Name                  : %s\n", NAME);
551             printf("Student ID                    : %s\n", ID);
552             printf("Quality Points Earned         : %.2f\n",          ↵
                  TOTAL_QUALITY_POINTS1);
553             printf("Total Quality Points Earned   : %.2f\n",          ↵
                  TOTAL_QUALITY_POINTS1);
554             printf("Credit Hours Earned           : %d\n",            ↵
                  TOTAL_CREDIT_HOURSS1);
555             printf("Total Credit Hours Earned     : %d\n",            ↵
                  TOTAL_CREDIT_HOURSS1);
556             printf("Student's GPA                 : %.2f\n", GPAS1);
557             printf("Student's CGPA                : %.2f\n", CGPAS1);
558             printf                                                    ↵
                  ("========================================================= ↵
                  ==\n\n");
559         }
560
561         else if (SEMESTER == 2)
562         {
563             printf("\n\n");
564             printf                                                    ↵
                  ("========================================================= ↵
                  =\n");
565             printf("Student Name                  : %s\n", NAME);
```

```
566             printf("Student ID                      : %s\n", ID);
567             printf("Quality Points Earned          : %.2f\n",
                  TOTAL_QUALITY_POINTS2);
568             printf("Total Quality Points Earned    : %.2f\n",
                  TOTAL_QUALITY_POINTS1 + TOTAL_QUALITY_POINTS2);
569             printf("Credit Hours Earned            : %d\n",
                  TOTAL_CREDIT_HOURSS2);
570             printf("Total Credit Hours Earned      : %d\n",
                  TOTAL_CREDIT_HOURSS1 + TOTAL_CREDIT_HOURSS2);
571             printf("Student's GPA                  : %.2f\n", GPAS2);
572             printf("Student's CGPA                 : %.2f\n", CGPAS2);
573             printf
                  ("============================================================
                  ==\n\n");
574         }
575
576         else if (SEMESTER == 3)
577         {
578             printf("\n\n");
579             printf
                  ("============================================================
                  =\n");
580             printf("Student Name                   : %s\n", NAME);
581             printf("Student ID                     : %s\n", ID);
582             printf("Quality Points Earned          : %.2f\n",
                  TOTAL_QUALITY_POINTS3);
583             printf("Total Quality Points Earned    : %.2f\n",
                  TOTAL_QUALITY_POINTS1 + TOTAL_QUALITY_POINTS2 +
                  TOTAL_QUALITY_POINTS3);
584             printf("Credit Hours Earned            : %d\n",
                  TOTAL_CREDIT_HOURSS3);
585             printf("Total Credit Hours Earned      : %d\n",
                  TOTAL_CREDIT_HOURSS1 + TOTAL_CREDIT_HOURSS2 +
                  TOTAL_CREDIT_HOURSS3);
586             printf("Student's GPA                  : %.2f\n", GPAS3);
587             printf("Student's CGPA                 : %.2f\n", CGPAS3);
588             printf
                  ("============================================================
                  ==\n\n");
589         }
590
591         else
592         {
593             printf("Invalid semester");
594             printf("\n\n");
595         }
596     } while (SEMESTER != 0);
597     return 0;
598 }
```

```
599
600 void studentMalay()
601 {
602     printf("Masuk ID Murid untuk log masuk : ");
603     scanf("%s", ID);
604     do {
605         printf("Masuk sesi semester  ( 0 untuk keluar ) : ");
606         scanf("%d", &SEMESTER);
607         if (SEMESTER == 1)
608         {
609             printf("\n\n");
610             printf                                                    ↵
                  ("============================================================ ↵
                  =\n");
611             printf("Nama murid                      : %s\n", NAME);
612             printf("ID murid                        : %s\n", ID);
613             printf("Point Kualiti diperolehi        : %.2f",          ↵
                  TOTAL_QUALITY_POINTS1);
614             printf("Jumlah Point Kualiti diperolehi : %.2f",          ↵
                  TOTAL_QUALITY_POINTS1);
615             printf("Jam kredit diperolehi           : %d\n",          ↵
                  TOTAL_CREDIT_HOURSS1);
616             printf("Jumlah jam kredit diperolehi    : %d\n",          ↵
                  TOTAL_CREDIT_HOURSS1);
617             printf("GPA murid                       : %.2f\n", GPAS1);
618             printf("CGPA murid                      : %.2f\n", CGPAS1);
619             printf                                                    ↵
                  ("============================================================ ↵
                  ==\n\n");
620         }
621
622         else if (SEMESTER == 2)
623         {
624             printf("\n\n");
625             printf                                                    ↵
                  ("============================================================ ↵
                  =\n");
626             printf("Nama murid                      : %s\n", NAME);
627             printf("ID murid                        : %s\n", ID);
628             printf("Point Kualiti diperolehi        : %.2f",          ↵
                  TOTAL_QUALITY_POINTS2);
629             printf("Jumlah Point Kualiti diperolehi : %.2f",          ↵
                  TOTAL_QUALITY_POINTS1 + TOTAL_QUALITY_POINTS2);
630             printf("Jam kredit diperolehi           : %d\n",          ↵
                  TOTAL_CREDIT_HOURSS2);
631             printf("Jumlah jam kredit diperolehi    : %d\n",          ↵
                  TOTAL_CREDIT_HOURSS1 + TOTAL_CREDIT_HOURSS2);
632             printf("GPA murid                       : %.2f\n", GPAS2);
633             printf("CGPA murid                      : %.2f\n", CGPAS2);
```

```
634             printf
                  ("============================================================ 
                  ==\n\n");
635         }
636
637         else if (SEMESTER == 3)
638         {
639             printf("\n\n");
640             printf
                  ("============================================================ 
                  =\n");
641             printf("Nama murid                      : %s\n", NAME);
642             printf("ID murid                        : %s\n", ID);
643             printf("Point Kualiti diperolehi        : %.2f",
                  TOTAL_QUALITY_POINTS3);
644             printf("Jumlah Point Kualiti diperolehi : %.2f",
                  TOTAL_QUALITY_POINTS1 + TOTAL_QUALITY_POINTS2 +
                  TOTAL_QUALITY_POINTS3);
645             printf("Jam kredit diperolehi           : %d\n",
                  TOTAL_CREDIT_HOURSS3);
646             printf("Jumlah jam kredit diperolehi    : %d\n",
                  TOTAL_CREDIT_HOURSS1 + TOTAL_CREDIT_HOURSS2 +
                  TOTAL_CREDIT_HOURSS3);
647             printf("GPA murid                       : %.2f\n", GPAS3);
648             printf("CGPA murid                      : %.2f\n", CGPAS3);
649             printf
                  ("============================================================ 
                  ==\n\n");
650         }
651
652         else {
653             printf("\nSesi semeter tidak sah\n");
654             printf("\n\n");
655         }
656
657     } while (SEMESTER != 0);
658     return 0;
659 }
660
661 void invalid()
662 {
663     printf("\n\n");
664     printf("Invalid option / Kesalahan pilihan\a\a\a\a");
665     return 0;
666 }
667
668 void displayGradeScale()
669 {
670     printf("Grade Scale :\n");
```

```
671     printf("A  (80-100) - Outstanding     (4.00)\n");
672     printf("A- (75-79)  - Excellent       (3.75)\n");
673     printf("B+ (70-74)  - Very Good       (3.50)\n");
674     printf("B  (65-69)  - Good            (3.00)\n");
675     printf("B- (60-64)  - Above Average   (2.75)\n");
676     printf("C+ (55-59)  - Average         (2.50)\n");
677     printf("C  (50-54)  - Below Average   (2.00)\n");
678     printf("F  (0-49)   - Weak            (0.00)\n");
679     return 0;
680 }
681
```